

MOHAMED BEN HASSEN

AI ENGINEER | FULL-STACK DEVELOPER | CLOUD & INFRASTRUCTURE

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PROFESSIONAL SUMMARY

AI Engineer and Developer with hands-on experience across cloud infrastructure, AI systems, automation, full-stack development, and system operations. Experienced with Azure virtual machines, networking, secure access, firewalling, load balancing, backup/restore, containerized application deployment, and Microsoft cloud services. Currently developing a company-wide digital platform for iPrint International with AI-assisted email workflows, automation, structured document generation, role-based access, and operational tracking. Strong focus on building practical systems that connect cloud, AI, automation, and product execution.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, C++, R, HTML, CSS

AI / ML: TensorFlow, Scikit-Learn, CNNs, YOLOv8, OpenCV, OpenPose, recommender systems, Pandas, NumPy, Matplotlib

Generative AI / Agents: AI agents, RAG, LangChain, LangGraph, MCP, LLM workflows, prompt engineering, structured extraction

Automation / Integration: n8n, Microsoft Power Automate, Microsoft Graph API, REST APIs, webhooks, workflow automation

Cloud / Infrastructure: Microsoft Azure, Azure VMs, VNets, Subnets, NSGs, Azure Firewall, Bastion, Load Balancer, AKS, snapshots/restore, Microsoft Entra ID

Databases / Data: SQL, NoSQL, PostgreSQL, MySQL, MongoDB, Cosmos DB, Supabase

MLOps / Backend: Docker, Docker Compose, Kubernetes, MLflow, Apache Airflow, Neptune, Flask, FastAPI, Node.js

Frontend / Product: React, Next.js, Vite, Framer Motion, responsive UI/UX, Git/GitHub, Vercel, GitHub Pages

PROFESSIONAL EXPERIENCE

iPrint International - AI Engineer & Developer (Project-Based) | Doha, Qatar | Aug 15, 2026 - Present

- Developing a custom company-wide digital platform to centralize workflows across Client Servicing, Finance, Traffic, Production, and other departments.
- Designing role-based workflows, structured document generation, Excel generation, job-brief mapping, tracking dashboards, and reusable data-entry flows to reduce manual work.
- Building AI-assisted email workflows for classification, summarization, structured extraction, request routing, and mapping email content into operational records and documents.
- Integrating Microsoft Graph, Microsoft Entra ID, Power Automate, APIs/webhooks, and database-backed workflow logic for email and process automation.

VisioRide - Technology Department | Qatar | Sep 9, 2026 - Present

- Supporting software development, AI integration, automation, security improvement, Git/GitHub workflows, and production-focused web platform enhancements.
- Contributing to technical planning and implementation for internal tools, intelligent digital features, and scalable platform improvements.

Qatar Mobility Innovations Center (QMIC) - AI Engineer Intern | Doha, Qatar | Jun 2024 - Aug 2024

- Researched and curated satellite-imagery datasets, converted annotations to YOLO format, trained and fine-tuned YOLOv8 vessel detection/classification models, and resolved framework/version issues.
- Integrated segmentation-based weights into a self-driving perception workflow; refined data/annotations and improved model performance by more than 4%.
- Performed model testing, output analysis, and statistical evaluation to compare iterations and validate improvements.

EDUCATION

Bahcesehir University | Istanbul, Turkiye

Bachelor of Science - Artificial Intelligence Engineering | Master's Degree - Artificial Intelligence Engineering | GPA: 3.17 / 4.00

SELECTED CERTIFICATIONS

Microsoft Azure Cloud Services • Applied Machine Learning • Machine Learning with Python • DevOps, DataOps & MLOps • MLOps Tools: MLflow & Hugging Face • Python Essentials for MLOps • Neural Networks and Deep Learning • Improving Deep Neural Networks • 9 Kaggle micro-certificates.

LANGUAGES

Arabic - Native | English - C2 / Proficient | French - B2 / Upper Intermediate | Turkish - B1 / Intermediate

TECHNICAL PROJECTS

Azure Infrastructure, Secure Networking & Cloud Deployment

- Provisioned and administered Azure Linux virtual machines using SSH key authentication, managed compute/storage settings, and configured secure administrative access.
- Designed lab-scale network topologies using VNets, subnets, NSGs, route tables, Azure Firewall, and Bastion; validated controlled traffic flows between protected subnets and services.
- Built and tested an Internal Load Balancer environment and configured backend VMs, health probes, load-balancing rules, and private network connectivity.
- Implemented VM backup/restore exercises using snapshots and restored disks/VMs to validate recovery procedures.
- Built and deployed a containerized Flask application on Azure using Docker, AKS, and Cosmos DB, covering application packaging, orchestration, and cloud data integration.

Enterprise Network Design & Simulation

- Designed and configured a small-company network in Cisco Packet Tracer with routed internal and external network segments, switches, two internal clients, a web server, a DNS server, and an external client.
- Configured static IP addressing, default gateways, DNS name resolution, and HTTP services; validated end-to-end connectivity using ICMP tests and browser access to the hosted company website from both internal and external clients.

Qatar National Datathon

- Collaborated on an OpenAI-powered analytics website with voice recognition and data visualizations, allowing conversational exploration of NPC datasets and labor-force insights.
- Built a predictive model to assess STEM-enrollment growth toward Qatar Vision 2030 targets and contributed data-driven recommendations for technology-sector talent development.

News Recommender System

- Designed and evaluated content-based, collaborative, and hybrid recommenders on Microsoft MIND-small using TF-IDF user profiles and an implicit user-item matrix with Truncated SVD.
- Experimented with ML, deep-learning-integrated, and LLM-assisted approaches; content-based filtering performed best overall (HR@10 0.673, NDCG@10 0.357, Precision@10 0.078).

Deep Learning Systems: CNN Robustness & Failure Modes

- Built a fixed CNN and ran controlled one-variable-at-a-time experiments on activation functions, dataset size, and learning rate to study representation, generalization, and optimization failure modes.
- Compared ReLU vs. tanh (~91.5% vs. ~91.8% test accuracy); tested 100%/25%/10%/5% training data; best tested learning rate was 1e-3 with 91.83% test accuracy.

OpenPose Gait Analysis & Joint-Angle Measurement

- Used pretrained CMU OpenPose to extract lower-body keypoints from walking sequences and calculate joint angles for gait analysis and abnormal-pattern monitoring.

Computer Vision & Image Processing

- AR billboard compositing: built a replacement pipeline using HSV/morphology and contours for region detection, four-corner ordering, gamma-based brightness matching, perspective homography, and alpha blending; batch-processed background/poster combinations.
- Multi-scale image enhancement: implemented 5-level Gaussian/Laplacian pyramids with weighted reconstruction; processed 13 images with level weights [1.6, 1.4, 1.2, 1.1, 1.0].

Gold Bakery - Luxury E-commerce Frontend

- Designed and developed a responsive storefront with reusable React components, product/order flows, smooth scrolling, and motion-driven interactions. Live: gold-bakery.vercel.app

Personal Portfolio Website

- Built and deployed an interactive portfolio showcasing AI, cloud/MLOps, computer vision, and software-engineering projects. Live: mohamedbenhasse.github.io/mohamed-portfolio/

ML Lifecycle Management System

- Developed an ML lifecycle workflow using MLflow and Apache Airflow for experiment tracking, model management, pipeline orchestration, deployment, monitoring, and documentation, with TensorFlow and Scikit-Learn models.

CS188 Pacman Search & Adversarial AI

- Implemented classical search and adversarial decision algorithms in the Pacman environment, focusing on decision quality and search efficiency.